

IS (5) : Rivest-Shamir-Adleman (RSA)

CODE

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import random
import os

def gcd(a, b):
    while b:
        a, b = b, a % b
    return a

def mod_inverse(e, phi):
    for d in range(3, phi, 2): # Start from 3 and increment by 2 (only odd numbers)
        if (e * d) % phi == 1:
            return d
    return None

def is_prime(n):
    if n < 2:
        return False
    for i in range(2, int(n ** 0.5) + 1):
        if n % i == 0:
            return False
    return True

def generate_prime(bits=8):
    while True:
        num = random.randint(2**(bits-1), 2**bits - 1)
        if is_prime(num):
            return num

def generate_keypair():
    p = generate_prime()
    q = generate_prime()
    while q == p:
        q = generate_prime()
```

```

n = p * q
phi = (p - 1) * (q - 1)
e = 3
while gcd(e, phi) != 1:
    e += 2 # Ensure e is odd and coprime to phi

d = mod_inverse(e, phi)
return ((e, n), (d, n))

def encrypt(text, public_key):
    e, n = public_key
    return [pow(ord(char), e, n) for char in text]

def decrypt(ciphertext, private_key):
    d, n = private_key
    return ''.join(chr(pow(char, d, n)) for char in ciphertext)

def encrypt_file(file_path, encrypted_path, public_key):
    with open(file_path, 'r') as file:
        plaintext = file.read()

    encrypted_data = encrypt(plaintext, public_key)
    with open(encrypted_path, 'w') as file:
        file.write(','.join(map(str, encrypted_data)))

def decrypt_file(encrypted_path, decrypted_path, private_key):
    with open(encrypted_path, 'r') as file:
        encrypted_data = list(map(int, file.read().split(',')))

    decrypted_text = decrypt(encrypted_data, private_key)
    with open(decrypted_path, 'w') as file:
        file.write(decrypted_text)

if __name__ == "__main__":
    file_input = "./input.txt"
    file_encrypted = "./encrypted.txt"

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file_decrypted = "./decrypted.txt"

public_key, private_key = generate_keypair()

encrypt_file(file_input, file_encrypted, public_key)
print("File encrypted successfully.")

decrypt_file(file_encrypted, file_decrypted, private_key)
print("File decrypted successfully.")
```

OUTPUT

File encrypted successfully.
File decrypted successfully.

Input Text:

Jim liked driving around town with his hazard lights on.

Encrypted Text:

39326,49520,35213,30619,44026,49520,38678,38695,43781,30619,43781,1830
7,49520,209,49520,25728,3679,30619,21023,18307,42140,13249,25728,43781,
30619,28222,42140,18037,25728,30619,18037,49520,28222,49755,30619,4975
5,49520,30233,30619,49755,21023,13041,21023,18307,43781,30619,44026,49
520,3679,49755,28222,30233,30619,42140,25728,23368

Decrypted Text:

Jim liked driving around town with his hazard lights on.